

RAHUL RAMESH OMALUR

Atlanta, Georgia · U.S. Citizen

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Education

Indraprastha Institute of Information Technology (IIITD)

Nov. 2022 – May. 2026

Bachelor of Technology in Computer Science and Artificial Intelligence

New Delhi, Delhi

CGPA: 8.69

Relevant Coursework

LLMs · Deep Learning · NLP · Machine Learning · Computer Vision · MLOps · Software Development · OS · DBMS

Experience

TCS Research & Innovation

May 2025 – July 2025

Machine Learning Intern

Thane, Maharashtra

- Engineered a **92-feature pipeline** (50 metadata + 42 ECG-derived features) over **318 patient records**, applying **PCA** and imbalance handling on a skewed CAD-to-normal ratio (**197:22**).
- Benchmarked **4 classifiers** across **19 experiments**; best RFC (ECG+Meta) achieved **0.87 accuracy** and **0.93 F1**, with HistGBC attaining highest minority-class sensitivity (**G-score: 0.61**).
- Surfaced a clinical trade-off between overall accuracy and CAD-recall (F1-flip up to **0.38** vs. near-zero for SVM/RFC), informing model selection for high false-negative-risk deployment.

Prescription OCR – VLM Benchmarking

Jan 2026 – May 2026

Research Project

IIITD

- Benchmarked **10+ VLMs** (Qwen3-VL, InternVL3, PaliGemma) across **3 architectural paradigms** for zero-shot antibiotic entity recognition from handwritten prescriptions.
- Curated and augmented a **1,115-image benchmark** with domain-relevant transforms to simulate real-world clinical image degradation.
- Introduced **WanIOU**, a weighted set-level metric immune to shortcut inflation; best pipeline achieved **WanIOU 0.2970**, vs. GLM-OCR (SOTA) collapsing to **0.04** despite **44.93% binary accuracy**.

Projects

Probabilistic Pundit | Multi-Agent LLMs, Groq, TOPSIS, LangChain

Aug 2025 – Dec 2025

- Built a hierarchical multi-agent LLM framework for FPL squad selection, fusing structured player stats with unstructured signals scraped from Reddit and official press sources.
- Designed a **TOPSIS**-based candidate filter (k=60) achieving RMSE ≈ 56 against global optimal squads, reducing the 600+ player universe while preserving **90%+** of theoretical point potential.
- Outperformed single-LLM baselines (Gemini 2.5 Flash, Kimi-k2, Llama-4) and a deterministic LP solver, achieving best RMSE of **105.56** with near-zero team value drift (**-0.16M**).

QUADRA – Intent-Aware Counterspeech Generation | HuggingFace, PyTorch

Jan 2025 – May 2025

- Designed a modular framework with a shared **HateBERT** encoder and **4 intent-specific BART** decoders trained on **9,532** examples.
- Proposed **DialoRank**, a zero-shot reranker for intent selection achieving **BERTScore F1 of 0.871**.
- Outperformed DialoGPT by **+12.1 ROUGE-1** and **+7.2 BERTScore F1**; improved intent accuracy from **0.70 (QUARC)** to **0.75** via cross-modal fusion.

SentinelAI – Intrusion Detection System | OpenCV, PyTorch, ONNX, Flask, Next.js

Jan 2026 – May 2026

- Built a multi-factor IDS fusing face recognition (YuNet+SFace), ByteTrack person tracking, and ST-GCN action recognition into a real-time threat-scoring pipeline.
- Attained **85.06% identity accuracy** while reducing known-subject latency from **51.7ms** to **23.9ms**.
- Deployed on Raspberry Pi 4 via **ONNX Runtime** quantization, cutting memory by **40–55%** (606MB RSS) and sustaining **1.2–1.4 FPS** for real-time local intrusion detection.

Technical Skills

Languages: Python, SQL, Java, C, PHP

AI & Machine Learning: PyTorch, TensorFlow, LangChain, ONNX, DSPy, OpenCV, NeuroKit

Cloud & Tools: AWS SageMaker, GCP, Groq, Runpod, Hugging Face, n8n, MCP

Leadership / Extracurricular

AudioBytes

Nov 2025 – May 2026

Co-ordinator

- Organized musical and technical events for college fests, coordinating logistics across teams.
- Led the college band for various inter-college competitions and festivals.